Chemistry Topic 5 — Carbon, Organic Compounds, Polymers and Fibres

Covers syllabus (click to expand)

Unique Bonding of Carbon (Catenation, Tetravalency, Hybridization) | Saturated & Unsaturated Hydrocarbons (Alkanes, Alkenes, Alkynes, Aromatics) | Master Allotropes of Carbon (Diamond, Graphite, Fullerenes, Carbon Nanotubes, Graphene, Black Gold) & Chronological Evolution | Polymer Classification (Addition vs Condensation, Linear vs Cross-linked) | Mechanical Behavior Classes (Elastomers, Fibres, Thermoplastics, Thermosetting Plastics) | Master Plastics & Resins (Bakelite, Teflon, PVC, LDPE, HDPE, Polystyrene, Melamine) | Natural, Regenerated & Synthetic Fibres (Rayon Cellulose Origin, Nylon-6,6, Terylene/Dacron, Kevlar Bulletproof Aramid) | Natural & Synthetic Rubbers (Isoprene Monomer, Sulphur Vulcanization, Neoprene, Buna-S, Buna-N) | Natural Biopolymers (Fibroin Silk, Keratin Wool, Cellulose, Starch)

Sources baked in: NCERT Class 10 (Carbon and its Compounds), Class 12 (Polymers; Chemistry in Everyday Life), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Rayon as regenerated cellulose (not starch), Neoprene as an elastomer (not thermoplastic), Kevlar-Fullerene-Graphene discovery chronology, Teflon non-stick coatings, Bakelite electrical switches, Isoprene in natural rubber, and Sulphur vulcanization. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Biodegradable Plastics & Biopolymers**	Mandatory phase-out of single-use petroleum-based plastics in India under Plastic Waste Management Rules. Accelerated adoption of **PLA (Polylactic Acid)** derived from fermented corn starch and **PHBV (Poly-\beta-hydroxybutyrate-co-\beta-hydroxyvalerate)**, which degrade into natural bio-assimilable CO_2 and H_2O via bacterial action within 180 days.
2024	**UPPCS Prelims 2024**	Direct question on **Rayon Origin**: Keyed that Rayon is a **regenerated cellulose fibre** obtained by chemically treating natural wood pulp or cotton linters. The option "modified starch" is an established distractor.
2022	**UPPCS Prelims 2022**	Polymer Class Matching: Identified **Neoprene** strictly as a synthetic **elastomer** (synthetic rubber), not a thermoplastic. Confirmed **Fibroin** is the natural structural protein of silk, not a synthetic polymer.
2020	**UPPCS Prelims 2020**	Tested the **Chronological Discovery Order of Carbon & Synthetic Materials**: **Kevlar (1965)** \to **Fullerenes (C_{60}, 1985)** \to **Carbon Nanotubes (1991)** \to **Graphene (2004)** \to **Black Gold (2019)**.

Year	Event	Core Concept & Must-Score Significance
2019	**UPPCS Prelims 2019**	Highlighted **Kevlar** as a high-strength aromatic polyamide (aramid fibre) utilized in bulletproof vests and lightweight armor due to its high tensile strength-to-weight ratio (5\times steel).

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master Commercial Polymer Classification (NCERT / Lucent Standard) | Polymer Trade Name | Monomer Name(s) & Chemical Formula | Polymerization Type & Linkage | Mechanical / Thermal Class | Core Real-World & Applications | |---|---|---|---|---| | **Low-Density Polyethene (LDPE)** | Ethylene (CH_2=CH_2) | Free-radical Addition; branched | Thermoplastic | Squeeze squeeze-bottles, packaging film, plastic bags, cable insulation | | **High-Density Polyethene (HDPE)** | Ethylene (CH_2=CH_2) | Ziegler-Natta Addition; linear | Thermoplastic | Rigid buckets, dustbins, industrial chemical drums, underground water pipes | | **Polyvinyl Chloride (PVC)** | Vinyl Chloride (CH_2=CHCl) | Addition | Thermoplastic | Water drainage pipes, electrical wire sheathing, raincoats, vinyl floor tiles | | **Teflon (PTFE)** | Tetrafluoroethylene (CF_2=CF_2) | Addition | Thermoplastic | Non-stick cooking cookware coatings, chemical-resistant gaskets, valve seats | | **Polystyrene** | Styrene (C_6H_5-CH=CH_2) | Addition | Thermoplastic | Expanded foam (Thermocol), packaging material, disposable drink cups | | **Bakelite** | Phenol (C_6H_5OH) + Formaldehyde (HCHO) | Condensation; -CH_2- cross-links | **Thermosetting Plastic** | Electrical wall switches, plugs, sockets, handles of frying pans, distributor caps | | **Melamine Resin** | Melamine + Formaldehyde | Condensation; cross-linked | **Thermosetting Plastic** | Unbreakable dinnerware, tableware crockery, decorative Formica laminates | | **Nylon-6,6** | Adipic acid + Hexamethylenediamine | Condensation; **Polyamide (-CONH-)** | **Fibre** | Toothbrush bristles, parachute fabrics, climbing ropes, tyre cords | | **Nylon-6** | Caprolactam (cyclic 6-carbon monomer) | Ring-opening Condensation; Polyamide | **Fibre** | Tyre cords, industrial fabrics, heavy-duty ropes | | **Terylene / Dacron** | Terephthalic acid + Ethylene glycol | Condensation; **Polyester (-COO-)** | **Fibre** | Crease-resistant clothing fabrics, safety belts, sails, PET plastic bottles | | **Kevlar** | 1,4-phenylenediamine + Terephthaloyl chloride | Condensation; **Aromatic Aramid** | **High-Modulus Fibre** | Bulletproof tactical vests, combat helmets, military armor, aerospace composites | | **Natural Rubber** | Isoprene (2-methyl-1,3-butadiene) | Addition; *cis*-1,4-polyisoprene | **Elastomer** | Surgical gloves, footwear soles, rubber bands (after vulcanization: vehicle tyres) | | **Neoprene** | Chloroprene (2-chloro-1,3-butadiene) | Addition; polychloroprene | **Elastomer** | Gasoline-resistant hoses, fuel gaskets, conveyor belts, diving wetsuits | | **Buna-S (SBR)** | 1,3-Butadiene + Styrene | Addition copolymer | **Elastomer** | Automobile tyres, footwear heels, conveyor belting | | **Buna-N (NBR)** | 1,3-Butadiene + Acrylonitrile | Addition copolymer | **Elastomer** | Fuel tank linings, industrial oil seals, chemical-resistant gloves | --- ### Table 2: Chronological Synthesis & Evolution of Carbon Allotropes & Synthetic Polymers | Material Name | Chemical Classification / Architecture | Year First Discovered / Synthesized | Key Discoverers / Institution | Definitive Examination Signature | |---|---|---|---|---| | **Vulcanized Rubber** | Cross-linked polyisoprene with sulphur | **1839** | Charles Goodyear (USA) | Disulphide bridges transform sticky latex into durable tyre rubber. | | **Bakelite** | Phenol-formaldehyde cross-linked resin | **1907** | Leo Baekeland (USA) | World's first fully synthetic plastic; thermosetting; electrical switches. | | **Nylon-6,6** | Aliphatic condensation polyamide | **1935** | Wallace Carothers (DuPont) | First 100%

synthetic textile fibre; polyamide linkage. | | **Teflon (PTFE)** | Perfluorinated addition polymer | **1938** | Roy Plunkett (DuPont) | Accidental discovery; non-stick, non-wetting, chemically inert up to 260^\circ\text{C}. | | **Kevlar** | Para-aramid aromatic polyamide | **1965** | Stephanie Kwolek (DuPont) | 5\times strength of steel on equal weight basis; bulletproof vests. | | **Fullerene (C_{60})** | Truncated icosahedron carbon cage | **1985** | Kroto, Curl, and Smalley | 20 hexagons and 12 pentagons; Nobel Prize in Chemistry, 1996. | | **Carbon Nanotubes (CNTs)**| Cylindrical tubular graphene fullerene | **1991** | Sumio Iijima (NEC Japan) | Helical carbon cylinders with high aspect ratio and ballistic conductivity. | | **Graphene** | 2D monolayer honeycomb carbon sheet | **2004** | Andre Geim & Konstantin Novoselov | Exfoliated with scotch tape; single atom thick; Nobel Prize in Physics, 2010. | | **Black Gold** | Plasmonic nanostructured gold arrangement | **2019** | TIFR Mumbai (Vivek Polshettiwar) | Broad solar light absorption for seawater desalination and CO_2 capture. | --- ### Table 2B: Coal Classification & Carbon Allotrope Hardness Hierarchy | Coal / Carbon Variety | Carbon Content (% by wt) | Calorific Value / Specific Properties | Primary Uses / Key Insights | | :--- | :---: | :--- | :--- | | **Peat** | < 40\% | Lowest grade; high moisture; smoky flame | Precursor to coal formation; soil conditioner | | **Lignite** (Brown Coal) | 25\% - 35\% | High moisture (\sim 35\%); low heating value | Thermal power generation (Neyveli, Tamil Nadu) | | **Sub-Bituminous** | 35\% - 45\% | Intermediate rank; low sulfur | Industrial boilers & electricity generation | | **Bituminous** (Soft Coal) | 45\% - 86\% | Most abundant commercial coal; dense, dark | Coking coal for steel metallurgy; heating | | **Anthracite** (Hard Coal) | 86\% - 97\% | Highest carbon rank; smokeless, semi-metallic luster | Domestic heating, highest energy output per unit mass | > [!TIP] > **Carbon Allotrope Highlights & Lab-Grown Diamonds:** > - **Thermodynamic Stability:** **Graphite** is thermodynamically the most stable allotrope of carbon at standard temperature and pressure (\Delta H_f^\circ = 0\text{ kJ/mol}). > - **Diamond Hardness:** Hardness = 10 on Mohs scale; highest thermal conductivity of any natural material; cut by laser or etched by **Hydrofluoric acid (HF)**. > - **Lab-Grown Diamonds (LGDs):** Synthesized via **High Pressure High Temperature (HPHT)** or **Chemical Vapour Deposition (CVD)** using **graphite or small diamond fragments as seeds**. > - **Bond Order of CO Group:** Carbon monoxide has 10 valence electrons; Molecular orbital configuration yields \text{Bond Order} = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(8 - 2) = \mathbf{3}. > - **Dry Ice:** Solid carbon dioxide (CO_2) sublimes at -78.5^\circ\text{C} without liquid transition; non-toxic cooling agent.

Detailed Theory Notes

5.1 Carbon Bonding, Hybridization & Hydrocarbon Families

1. Catenation and Carbon's Unique Position

Carbon occupies Group 14 and Period 2 of the Periodic Table. Its electronic configuration is 1s^2 2s^2 2p^2. - **Catenation**: The unique property of elements to form covalent bonds with atoms of the same element, generating extended chains, branched networks, and rings. \text{Catenation Capability: } C \gg Si > Ge \approx Sn > Pb The exceptional catenation of carbon stems from its small atomic radius (77\text{ pm}), which allows effective nuclear overlap and generates a high bond energy (348\text{ kJ/mol}). Silicon forms Si-Si bonds, but their lower energy (222\text{ kJ/mol}) and reactivity toward oxygen make long silicon chains unstable under ambient conditions.

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      H   H
      |   |
  H - C - C - H   (Ethane: sp³, single bond, σ only, bond angle 109.5°)
      |   |
      H   H

      H   H
       \ /
        C = C     (Ethene: sp², double bond, 1σ + 1π, bond angle 120°)
       / \
      H   H

  H - C ≡ C - H   (Ethyne: sp, triple bond, 1σ + 2π, bond angle 180°)
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2. Master Hydrocarbon Taxonomy

1. **Alkanes (C_n H_{2n+2})**: - Saturated aliphatic hydrocarbons. Every carbon is sp^3 hybridized with single \sigma-bonds. - Chemically unreactive at room temperature; undergo free-radical substitution reactions under UV light (e.g., chlorination: CH_4 + Cl_2 \xrightarrow{h\nu} CH_3Cl + HCl). - Methane (CH_4) accounts for >85\% of natural gas and is generated by methanogenic archaebacteria in oxygen-deprived environments (paddy fields, wetlands, cattle rumens).
2. **Alkenes (C_n H_{2n})**: - Unsaturated hydrocarbons containing a C=C double bond (1\sigma + 1\pi). - The \pi-bond is formed by sideways lateral overlap of unhybridized p-orbitals and is susceptible to electrophilic addition reactions (hydrogenation, halogenation). - Ethene (C_2H_4) is an endogenous plant hormone promoting fruit ripening and the monomer for polyethylene.
3. **Alkynes (C_n H_{2n-2})**: - Unsaturated hydrocarbons with a C \equiv C triple bond (1\sigma + 2\pi). - Acetylene (C_2H_2) burns in pure oxygen to yield an intensely hot flame (\approx 3200^\circ\text{C}), utilized in oxy-acetylene welding. Terminal alkynes have weakly acidic acetylenic protons that react with heavy metal salts (AgNO_3, Cu_2Cl_2) to form insoluble explosive acetylides.
4. **Aromatic Hydrocarbons (Arenes)**: - Planar cyclic conjugated compounds containing (4n+2) delocalized \pi electrons (Hückel's Rule). - Benzene (C_6H_6) exhibits extraordinary thermodynamic stability due to resonance energy (\approx 150\text{ kJ/mol}). It undergoes electrophilic aromatic substitution rather than addition reactions, preserving the intact aromatic \pi-system.

5.2 Allotropes of Carbon & Advanced Nanostructures

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                                  ┌── Diamond: 3D sp³, tetrahedral, hardest natural substance,
                                  ├── Graphite: 2D sp², hexagonal sheets, electrical conductor,
Carbon Allotropes (Pure Carbon) ──┼── Fullerenes (C₆₀): 0D cage, 20 hexagons + 12 pentagons ("b
                                  ├── Carbon Nanotubes (CNTs): 1D tubules, high tensile strengt
                                  └── Graphene: 2D monolayer, single atom thick, honeycomb latt
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1. Fullerenes (C_{60})

- Discovered in 1985 by laser vaporization of graphite by Kroto, Curl, and Smalley.
- Consists of 60 equivalent carbon atoms arranged in a spherical cage resembling a soccer ball.

- Geometry: Truncated icosahedron containing **20 hexagonal rings and 12 pentagonal rings**.
- *Isolated Pentagon Rule (IPR)*: No two pentagons share a common edge; every pentagon is entirely surrounded by five hexagons to minimize steric ring strain.
- Each carbon atom is sp^2 hybridized with slight pyramidal distortion to accommodate spherical curvature.

2. Graphene

- A single isolated two-dimensional layer of carbon atoms arranged in a planar hexagonal honeycomb lattice.
- Electrons move across the lattice as massless Dirac fermions with Fermi velocities near 10^6\text{ m/s}, enabling ballistic electronic transport.
- Extreme thermal conductivity (\approx 5000\text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}), Young's modulus of 1\text{ TPa}, and intrinsic tensile strength exceeding 130\text{ GPa} (>100\times stronger than structural steel).

5.3 Polymer Science: Reaction Mechanisms & Mechanical Classes

1. Polymerization Regimes

- **Addition Polymerization**: Monomers containing multiple bonds join without the loss of any small molecules. The empirical formula of the resulting polymer is identical to that of the starting monomer (e.g., n CH_2=CH_2 \to [-CH_2-CH_2-]_n).
- **Condensation Polymerization**: Intermolecular reactions between bi-functional or poly-functional monomers accompanied by the elimination of small byproduct molecules (H_2O, HCl, NH_3). The elemental composition of the polymer differs from that of the monomers (e.g., Nylon-6,6, Terylene, Bakelite).

2. Master Mechanical Classification

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[ Elastomers ]           [ Thermoplastics ]      [ Fibres ]              [ Thermosetting Pla
Weakest forces (van der  Intermediate forces     Strongest forces        Permanently cross-l
Waals); coiled chains;   (dipole/dispersion);    (H-bonds); high tensile 3D networks; non-me
elastic stretch.         reversibly meltable.    strength, minimal stretch non-recyclable.
(Rubber, Neoprene)       (PE, PVC, Polystyrene)  (Nylon, Terylene, Kevlar)(Bakelite, Melamine
```

5.4 Master Plastics, Resins & Synthetic Fibres

1. Bakelite (Phenol-Formaldehyde Condensation Resin)

- Phenol reacts with formaldehyde in the presence of an acid (H^+) or alkaline (OH^-) catalyst via electrophilic aromatic substitution to form ortho- and para-hydroxymethylphenols.
- Linear condensation yields **Novolac**, a fusible soluble polymer used in protective varnishes.
- Heating Novolac with hexamethylenetetramine (source of additional formaldehyde) forms methylene bridges (-CH_2-) that cross-link the chains into **Bakelite**, a rigid, infusible, thermosetting three-dimensional polymer network.

- Standard industrial application: **Electrical wall switchboards, plugs, appliance sockets, and heat-resistant cookware handles**.

2. Teflon (Polytetrafluoroethylene, PTFE)

- Polymerized from tetrafluoroethylene (CF_2=CF_2) using ammonium persulphate catalyst at elevated pressures: n(CF_2=CF_2) \to [-CF_2-CF_2-]_n
- The high C-F bond dissociation energy (485\text{ kJ/mol}) and dense protective fluorine sheath render Teflon chemically inert against boiling aqua regia, strong mineral acids, and oxidizing agents up to 260^\circ\text{C}.
- Exhibits the third-lowest coefficient of friction of any known solid material, making it the standard **non-stick coating on domestic kitchenware** and self-lubricating bearings.

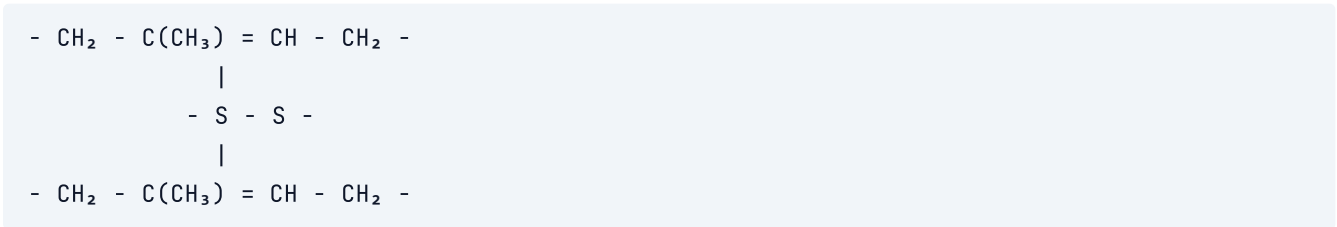
3. Kevlar (Para-Aramid)

- Synthesized by low-temperature polycondensation of **1,4-phenylenediamine** and **terephthaloyl chloride** in an anhydrous amide solvent: n H_2N-C_6H_4-NH_2 + n ClCO-C_6H_4-COCl \to [-HN-C_6H_4-NH-CO-C_6H_4-CO-]_n + 2n HCl
- The polymer chains form a rigid, rod-like conformation that self-assembles into liquid crystalline solutions. Upon spinning, the chains align parallel to the fiber axis, locked by dense intermolecular hydrogen bonds between amide groups and \pi-\pi stacking between aromatic rings.
- Provides exceptional tensile strength-to-weight ratio (\approx 5\times structural steel), high cut resistance, and dimensional stability up to 450^\circ\text{C}; standard material for **bulletproof vests, military helmets, and aerospace composites**.

5.5 Rubber Chemistry & Natural Biopolymers

1. Natural Rubber & Sulphur Vulcanization

- Natural rubber is a natural elastomer harvested as a colloidal suspension (latex) from *Hevea brasiliensis*. Chemically, it is strictly **cis-1,4-polyisoprene**: [-CH_2-C(CH_3)=CH-CH_2-]_n
- In the raw state, individual polyisoprene chains are held only by weak van der Waals interactions. Under mechanical stress, the chains slide past each other, leading to permanent deformation (tacky when warm, brittle when cold).
- **Vulcanization (Charles Goodyear, 1839)**:
- Heating raw rubber with **3–5\% elemental sulphur** at 140^\circ\text{C} in the presence of ZnO catalyst.
- Sulphur forms covalent **disulphide cross-links (-S-S-)** across the allylic carbon positions of adjacent chains:



- These chemical cross-links prevent chains from slipping permanently while allowing elastic stretching and snapping back, making the rubber durable and resilient for automobile tyres.

2. Synthetic Rubbers

- **Neoprene**: Prepared by addition polymerization of **Chloroprene (2-chloro-1,3-butadiene)**: n CH_2=C(Cl)-CH=CH_2 \to [-CH_2-C(Cl)=CH-CH_2-]_n The chlorine substituent provides polarity and chemical stability, making neoprene highly resistant to degradation by mineral oils, gasoline, and ozone.
- **Buna-S (SBR)**: Copolymer of 1,3-butadiene and styrene; standard material for automobile tread tyres.
- **Buna-N (NBR)**: Copolymer of 1,3-butadiene and acrylonitrile; used for fuel hoses and oil seals.

Bilingual Terminology -- Carbon, Organic Chemistry and Polymers

English Term	Hindi Term	Key Anchor
Organic Chemistry (कार्बनिक रसायन)	कार्बनिक रसायन	Study of carbon compounds; ~10 million known compounds
Carbon (कार्बन)	कार्बन	6th element; tetravalent; forms 4 bonds; backbone of life
Hydrocarbon (हाइड्रोकार्बन)	हाइड्रोकार्बन	Compound of only H and C; alkanes, alkenes, alkynes
Alkane (एल्केन)	एल्केन	CnH2n+2; saturated; single bonds; e.g., methane (CH4), ethane, propane
Alkene (एल्कीन)	एल्कीन	CnH2n; one double bond; unsaturated; e.g., ethylene (CH2=CH2)
Alkyne (एल्काइन)	एल्काइन	CnH2n-2; one triple bond; e.g., acetylene (CH = CH); used in welding
Benzene (बेंजीन)	बेंजीन	C6H6; aromatic ring; discovered by Faraday (1825); structure by Kekule (1865)
Polymer (बहुलक)	बहुलक	Large molecule made of repeating monomer units
Monomer (एकलक)	एकलक	Small repeating unit that combines to form polymer
Polymerisation (बहुलकीकरण)	बहुलकीकरण	Chemical process of joining monomers to form polymer
Plastic (प्लास्टिक)	प्लास्टिक	Synthetic polymer; thermoplastic (remeltable) or thermoset (permanent)
Rubber (रबड़)	रबड़	Natural = latex of Hevea brasiliensis; polymer of isoprene
Vulcanisation (वल्केनीकरण)	वल्केनीकरण	Heating rubber with sulphur; improves strength and elasticity
Cellulose (सेलुलोज)	सेलुलोज	Natural polymer of glucose; plant cell wall; most abundant organic compound

English Term	Hindi Term	Key Anchor
Nylon (नायलॉन)	नायलॉन	First synthetic fibre; polyamide; Wallace Carothers (Du Pont, 1935)
Isomers (समावयवी)	समावयवी	Same molecular formula, different structural formula; e.g., butane and isobutane

One-Liner Revision -- Carbon, Organic Chemistry and Polymers

#	Fast Fact
1	Carbon = tetravalent; can form 4 bonds; catenation (chain-forming) ability
2	Allotropes of carbon: Diamond (hardest), Graphite (soft, conductor), Fullerene (C60)
3	Diamond = 3D tetrahedral; hardest; electrical insulator
4	Graphite = layered hexagonal; soft; good electrical conductor; used as lubricant and in pencils
5	Fullerene (Buckminsterfullerene C60) = 60 carbon atoms in soccer-ball shape; discovered 1985
6	Methane (CH4) = simplest alkane; natural gas; greenhouse gas
7	Ethylene (C2H4) = simplest alkene; plant hormone (fruit ripening); makes polyethylene
8	Acetylene (C2H2) = simplest alkyne; used in oxyacetylene welding; very hot flame
9	Benzene (C6H6) = simplest aromatic compound; carcinogen; aromatic ring
10	Nylon = first synthetic fibre (1935, Du Pont); polyamide; used in parachutes, stockings
11	Polyester (PET) = from ethylene glycol + terephthalic acid; soft-drink bottles, clothing
12	Bakelite = first synthetic plastic (1907, Leo Baekeland); thermosetting; phenol + formaldehyde
13	PVC (polyvinyl chloride) = monomer = vinyl chloride; pipes, wires, credit cards
14	Polythene (polyethylene) = monomer = ethylene; plastic bags, bottles
15	Teflon (PTFE) = polytetrafluoroethylene; non-stick cookware; very low friction
16	Natural rubber = polymer of isoprene (C5H8)n; from Hevea brasiliensis
17	Vulcanisation = rubber + sulphur at 140 degC; makes rubber stronger and elastic (Goodyear, 1839)
18	Cellulose = polymer of beta-glucose; plant cell wall; not digestible by humans
19	Starch = polymer of alpha-glucose; energy storage in plants; digested by amylase
20	Proteins = polymer of amino acids; peptide bonds; 20 standard amino acids
21	DNA = polymer of nucleotides (deoxyribose + phosphate + base)
22	Soap = sodium/potassium salt of fatty acids; amphipathic molecule (hydrophilic + hydrophobic end)
23	Biodegradable plastics = break down by microorganisms; e.g., PLA (polylactic acid)

#	Fast Fact
24	Micro-plastics = plastic particles < 5 mm; accumulate in food chain; major environmental threat
25	Rayon = regenerated cellulose fibre; "artificial silk"; from wood pulp
26	LPG = mainly propane (C3H8) + butane (C4H10); ethyl mercaptan added as odorant
27	CNG = compressed natural gas (mainly methane CH4); cleaner than diesel
28	Ethanol (C2H5OH) = ethyl alcohol; from fermentation; used as fuel additive (E10 = 10% ethanol)
29	Methanol (CH3OH) = wood alcohol; poisonous; causes blindness and death
30	Chloroform (CHCl3) = trichloromethane; used as anaesthetic historically; now solvent